# STAFF REVIEW · H446 009 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q3

## Do Now — source answers

1. Unicode supports a much larger repertoire of characters from many writing systems. Standard ASCII has 128 codes and is centred on a limited character set. Unicode provides code points for many languages and symbols.

2. 5.5 Mantissa 01011000 = 0.6875 ; exponent 0011 = 3 ; $0.6875 \times 2^3 = 5.5$ .

3. $01101101_2$ 6 → 0110 and D → 1101 .

4. −10 −128 + 64 + 32 + 16 + 4 + 2 = −10 .

## I Do — source worked steps

1. Use the standard .NET Fiddle structure — CODE:
using System;

public class Program
{
public static void Main()
{
// code goes here
}
} — Main program instructions belong inside Main() .

2. Read the score as an integer — CODE:
Console.Write("Enter score: ");
int score =
int.Parse(Console.ReadLine()); — Console.ReadLine() gives us text; int.Parse(...) converts it to a whole number.

3. Test the highest threshold first — CODE:
if (score >= 70)
{
Console.WriteLine("Distinction");
}
else if (score >= 50)
{
Console.WriteLine("Pass");
}
else
{
Console.WriteLine("Not yet");
} — A score of 85 satisfies both >= 70 and >= 50 , so the higher threshold must be tested first.

4. Use values that expose mistakes — CODE:
85 → Distinction
70 → Distinction
50 → Pass
49 → Not yet

## We Do — guided CHECK

1. Build the missing decision
CODE:
```
Console.Write("Number: ");
int number = int.Parse(Console.ReadLine());

if (number % 2 == 0)
{
Console.WriteLine("Even");
}
else
{
Console.WriteLine("Odd");
}
```
C# % is the remainder operator; it matches OCR MOD .

2. Which condition comes first?
Test the highest threshold first so 25+ is not caught by the 15+ branch.

3. Write one Boolean condition
```
if (age >= 12 && age <= 17)
```

4. Name it and show it
Use the technical vocabulary: sequence, branching/selection, iteration.

## Page 3 — independent skill check, teacher-review model

1. Assuming age >= 0: if (age < 12) Child; else if (age < 18) Teen; else Adult. Boundary outputs: Child, Teen, Teen, Adult.

## You Do — source mark schemes

1. Positive, negative or zero
CODE:
```
using System;

public class Program
{
public static void Main()
{
Console.Write("Enter an integer: ");
int number = int.Parse(Console.ReadLine());

if (number > 0)
{
```

```
Console.WriteLine("Positive");
}
else if (number < 0)
{
Console.WriteLine("Negative");
}
else
{
Console.WriteLine("Zero");
}
}
}
```
Why it works: the first two conditions remove positive and negative values; the only remaining possibility is zero.

2. Ticket category
CODE:
```
using System;

public class Program
{
public static void Main()
{
Console.Write("Enter age: ");
int age = int.Parse(Console.ReadLine());

if (age <= 11)
{
Console.WriteLine("Child");
}
else if (age <= 17)
{
Console.WriteLine("Teen");
}
else
{
Console.WriteLine("Adult");
}
}
}
```
Why the order works: once ages 11 and below have been removed, age <= 17 means 12–17.

3. Delivery charge
CODE:
```
using System;

public class Program
{
```

```
public static void Main()
{
Console.Write("Order total: ");
double total = double.Parse(Console.ReadLine());

Console.Write("Member? yes/no: ");
string member = Console.ReadLine();

if (total >= 30 && member == "yes")
{
Console.WriteLine("Free delivery");
}
else
{
Console.WriteLine("Delivery charge applies");
}
}
}
```

Key idea: && means both Boolean expressions must be true.

4. Simple login

CODE:

```
using System;

public class Program
{
public static void Main()
{
Console.Write("Username: ");
string username = Console.ReadLine();

Console.Write("PIN: ");
int pin = int.Parse(Console.ReadLine());

if (username == "admin" && pin == 2468)
{
Console.WriteLine("Access granted");
}
else
{
Console.WriteLine("Access denied");
}
}
}
```

Key idea: equality uses == . The login succeeds only when both comparisons are true.

## Exit Ticket — source answer

1. True. In C#, == tests whether values are equal; = is assignment.